

■ Scientific Justification (required for all)

Does the dust around a changing nucleus keep up with it? An active galactic nucleus (AGN) can undergo large changes in continuum luminosity within years [1,2], in some cases accompanied by the appearance or disappearance of broad emission lines [3–5]. Its dust records these changes through delayed heating, sublimation, or replenishment. Studies of individual nuclei suggest that the inner dust boundary adjusts over several years [6,7]; what is missing is an ensemble test of how the response depends on the direction, amplitude, and timing of the transition.

We request **67.0 h of MIRI/MRS spectroscopy of 24 AGN at $z < 0.3$** caught at distinct transition stages. Combining these spectra with each AGN’s recorded activity history, we will ask how past activity shapes its present dust emission. ZTF/NEOWISE light curves and multi-epoch optical spectra trace nuclear variability, while SPHEREx’s four planned all-sky surveys will extend our sampling of the hot-dust emission through 2027. MRS will separate the mid-infrared continuum, both silicate features, and polycyclic aromatic hydrocarbon (PAH) bands, allowing us to probe cooler dust and constrain the host-galaxy contribution. The programme addresses two linked questions:

1. How long does warm-dust memory persist? Light-travel delays mean that dust can remain bright after the nucleus fades, or brighten only after a delay following an outburst [8,9]. Comparing historical WISE data with MIRI spectra tests how long the infrared emission continues to reflect an earlier luminosity state.

2. Does dust replenish after the nucleus fades? The simplest null model keeps the dust distribution fixed while its temperature responds to changing illumination. The continuum and silicate features test whether fading and recovering sources require additional dust at radii previously cleared by sublimation [6–9]. Rising sources provide a comparison for the response to increased luminosity.

WISE infrared echoes link changing-look transitions to intrinsic changes in nuclear power [10–13]. The sublimation radius scales roughly as $L^{1/2}$ [14–16], but observations suggest that the inner dust boundary takes several years to respond [6]. Mrk 590’s smaller dust radius after fading points to replenishment [7], whereas Mrk 817 shows temperature changes consistent with heating at a fixed boundary [17]. Establishing when delayed heating is sufficient and when the dust distribution must evolve will determine how reliably mid-infrared emission traces the current nuclear luminosity.

Why MRS? Spectroscopy separates the continuum, silicate features, PAH bands, and narrow lines blended in broadband imaging [18–20]. The 9.7 and 18 μm silicate features can appear in emission from warm dust or in absorption when cooler material obscures brighter emission. Their strengths and the continuum shape constrain temperature structure and optical depth [21,22], testing whether a fixed dust distribution heated by the recorded history can explain the spectrum. PAH emission commonly accompanies host-galaxy star formation; its spatial distribution and band ratios help constrain the host contribution while accounting for AGN processing [23].

MRS covers 4.9–27.9 μm [24], extending SPHEREx’s 0.75–5 μm spectra [17,25,26] to cooler dust. It includes the 18 μm feature beyond MIRI low-resolution spectroscopy’s $\sim 14 \mu\text{m}$ limit. Its integral-field data separate nuclear and surrounding host emission [23,27].

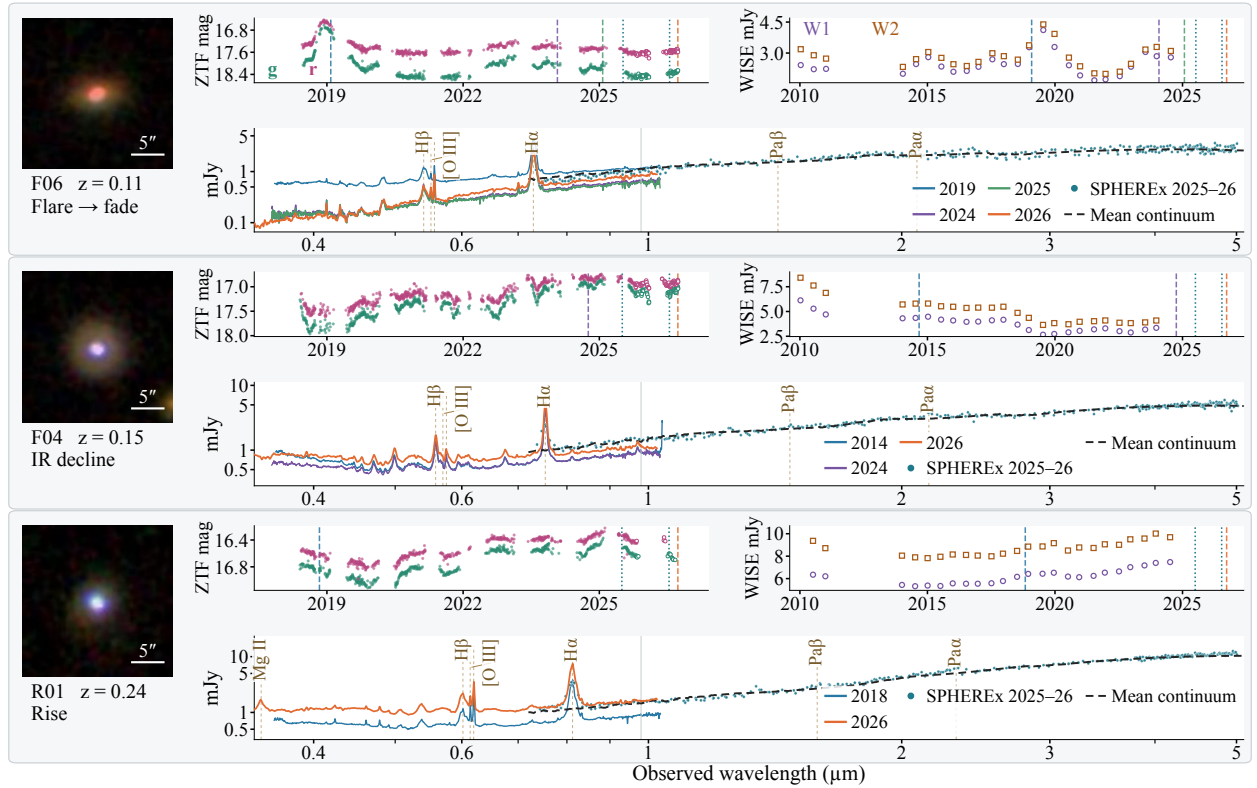


Figure 1: **Three examples of the rich histories assembled for the 24-target sample.** Each example combines ZTF and WISE light curves, archival and 2026 optical spectra and multiple SPHEREx visits, connecting nuclear activity to hot dust. **Top:** a 2018–2019 flare followed by fading; **middle:** an infrared decline with modest optical recovery; **bottom:** rising light curves and optical spectra. MIRI will test how much of this history the warm dust retains. SDSS images have 5'' bars. Open ZTF points are alerts through 2026 September; coloured ticks mark spectral epochs. Dashed curves show mean host+disc+blackbody fits to SPHEREx ($T_{\text{col}} \approx 1100\text{--}1200$ K, model-dependent); gold markers indicate emission lines. Wavelengths are observed-frame; spectra are in mJy.

Why this sample? Our 24 targets have diverse activity histories, each documented by ZTF and WISE light curves, SPHEREx coverage, and multi-epoch optical spectra, including observations obtained in 2026. These observations let us compare warm-dust spectra at different times after a luminosity change and test whether rising and fading sources differ at comparable current luminosity. Figure 1 illustrates the available observations for three targets.

We draw from catalogued changing-look AGN [28–31] and large-amplitude ZTF/WISE variables [5,32] spanning optical spectral types [29,33], using manifold learning to group similar light-curve shapes [34,35]. All lie at $z < 0.3$, keeping both silicate peaks within MRS coverage. W1/W2 monitoring spans ~ 10 years, and archival W3/W4 detections support exposure planning.

What we will measure and model. We will use wavelength-dependent point-spread-function (PSF) decomposition to separate nuclear and extended circumnuclear emission in each MRS cube [27] (Figure 2, lower left). The surrounding spectra constrain host light in the nuclear extraction. We will fit both components with the SPIRIT framework [19] (Figure 2, top), modelling dust emission, extinction, and PAH profiles. Comparison with CAFE [36] will assess how continuum placement and extinction assumptions affect the results.

We will measure three complementary sets of diagnostics: **(1) Dust luminosity and temperature:** the rest-frame 8–13 μm nuclear dust luminosity and continuum slopes over 5.5–14.5 and 14.5–20 μm , covered for all 24 targets. The luminosity retains broad silicate features, with PAHs and narrow lines removed; slopes use the fitted continuum. **(2) Dust properties and obscuration:** the 9.7 and 18 μm silicate strengths and profiles, which together constrain dust optical depth and temperature structure [21,22]. **(3) Host emission and excitation:** PAH fluxes and the 11.3/7.7 μm band ratio, compared between nuclear and surrounding spectra to assess host emission and AGN processing [23]. [Ne V] and [Ne VI] lines provide supporting excitation diagnostics.

Connecting one MRS epoch to the recorded history. Light-travel delays smooth the dust’s response to nuclear variability [8,37,38] (Figure 2, lower right). The existing optical and infrared monitoring therefore gives a single new MRS epoch a measured historical context. We will reconstruct the driving light curves from optical monitoring using Gaussian processes, propagating uncertainty across gaps and in the conversion to dust-heating luminosity. WISE light curves constrain the infrared response, while the 2010 W3/W4 photometry supplies an earlier warm-dust measurement. Each model predicts the spectrum at the actual JWST date, incorporating optical photometry near the visit and the same host component at both infrared epochs.

Testing delayed heating against dust evolution. We will compare two physical hypotheses. **Fixed dust distributions:** models with dust in clouds or in a disc and wind [39–41], heated by the recorded luminosity history with light-travel delays [8,9]. **Evolving dust distributions:** models that also allow destruction or replenishment [6,7,26], including changes in the dust supplied by a wind [42]. Both families allow for uncertainty in grain sizes and composition [43].

The optical/infrared history and archival warm-dust photometry will constrain the allowed distributions before we test their predicted MRS luminosity and spectral shape. We will assess evidence for evolution through the joint continuum and silicate fit, accounting for model uncertainty, measurement noise, host subtraction, and correlated calibration errors [27,44]. Simulated observations of fixed and evolving distributions, fitted with both model families, will calibrate the improvement required to favour evolution and determine which replenishment signals can be recovered.

Across the 24 targets, we will test whether warm-dust excesses or deficits relative to fixed-distribution predictions depend on whether the nucleus is rising or fading and on time since its luminosity changed, allowing for differences between sources [45,46]. This comparison will constrain how long warm dust retains a past luminosity state. Where replenishment is distinguishable from delayed heating, we will constrain its timescale and location; where fixed distributions suffice, we will place model-dependent limits on replenishment.

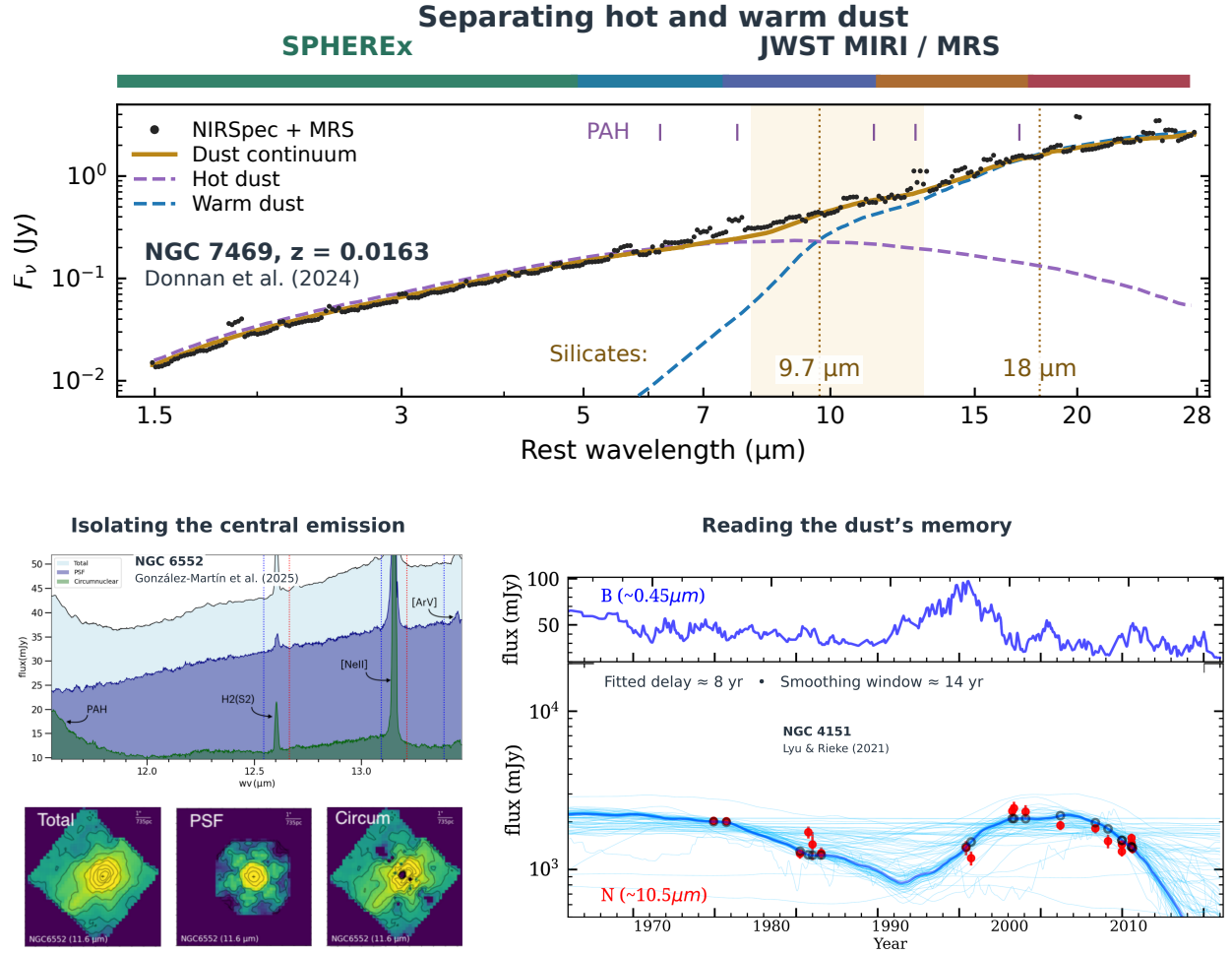


Figure 2: The spectral, spatial, and temporal ingredients of the experiment, each already shown to work in a single AGN. Top—Spectral: measuring the dust spectrum. The NIRSpec+MIRI/MRS spectrum of NGC 7469 (digitized from [19, Fig. 8]) separates cleanly into hot and warm dust components. The shaded 8–13 μm band provides the dust luminosity once PAH and line emission are subtracted, and the 9.7 and 18 μm silicate features add a spectral-shape diagnostic that imaging cannot supply. Bars mark MRS and, for comparison, SPHEREx coverage at $z = 0.0163$. **Lower left—Spatial: isolating the nucleus.** Spectra and 11.6 μm maps of NGC 6552 [27, Fig. 4] decompose the emission into nuclear (PSF) and circumnuclear components, the step that decides whether a nuclear warm excess survives host subtraction. **Lower right—Temporal: reading the dust’s memory.** In NGC 4151 the 10.5 μm emission (red points; light-blue curves are fitted responses) follows the optical light curve (blue) with an ~ 8 yr delay and additional smoothing [37, Fig. 15], preserving a record of earlier nuclear activity. Wavelengths are rest-frame in the top panel and observed-frame at lower left; the lower-right panel omits longer-wavelength data and ancillary fit labels. Our programme combines these measurements across 24 nuclei with recorded luminosity changes.

■ Description of Observations (required for all GO and Survey)

Spectral coverage and sensitivity. Each target will receive one MIRI/MRS science visit and a dedicated background visit, scheduled consecutively with identical integration parameters [47]. All three grating settings provide 4.9–27.9 μm coverage, including both silicate peaks for all 24 targets, with four-point dithers in each setting.

To measure the continuum shape and broad silicate profiles, we target S/N of 50 per $R = 100$ bin near rest 5 and 12 μm , and 30 at 18 μm (2% and 3.3% statistical errors, respectively). Exposure planning uses archival total-source W3/W4 fluxes spanning 1.9–48.4 and 6.4–82.6 mJy. APT specifies FULL/FASTR1 integrations, with extra setting-B exposure for the W3-faint J083826.50+371906.7. Equal-depth background visits measure the sky separately from the extended host emission [47]. Science-plus-sky integration totals 17.00 h; **the 67.0 h request** includes APT 2026.5.1 slew, acquisition, and exposure/setup overheads across 48 visits (24 science and 24 background). This is a Medium programme.

For J083826.50+371906.7 at $z = 0.2111$, ETC/Pandeia 2026.7 predicts S/N of 55/225/61 per $R = 100$ bin at rest 5/12/18 μm . These estimates assume a point source with $F_\nu \propto \lambda^2$, 0.95 mJy in W3 (half the archival total), the adopted sky spectrum, equal-depth sky exposures, and extraction radii of 0.3"/0.6"/0.9". They establish statistical sensitivity under the adopted nuclear-flux model; host and calibration uncertainties enter the science inference separately. Acquisition uses F560W for the faintest sources and the neutral-density filter for brighter nuclei, with readouts yielding S/N above 50 without saturation over the adopted flux ranges.

Separating nuclear dust from the host. Each MRS cube measures the nucleus and surrounding host together. The PSF FWHM grows from approximately 0.3" at 5 μm to 1" at 25 μm , sampling roughly 0.1–4.5 kpc across the sample; channel fields span 3.2" $\times$ 3.7" to 6.6" $\times$ 7.7" [24]. Wavelength-dependent PSF fitting [48] separates unresolved and resolved emission [27] (Figure 2, lower left). Surrounding spectra constrain host templates with allowance for spatial variation. Joint modelling separates continuum, silicate, PAH, and H₂ contributions [19,49], allowing for AGN processing of PAHs [23]. Alternative extractions and host templates [20] bound the nuclear flux, including compact host light. [Ne VI] 7.7 and [Ne V] 14.3 μm are accessible for all targets [50]; the nominal range also includes [Ne V] 24.3 μm for 13 and [O IV] 25.9 μm for nine. These lines support the excitation analysis without assuming an instantaneous response to nuclear power.

We will integrate observed-frame MRS spectra through the WISE responses, using the measured spectral shape to calculate synthetic W3/W4 fluxes in the WISE calibration convention. Comparison with 2010 requires matching the WISE spatial response: constant host light cancels in a flux difference only for the same spatial contribution. WISE imaging and aperture photometry constrain emission beyond the MRS field; optical and 2MASS profiles guide stellar-host morphology. Models account for spectral coverage gaps and host light within the SPHEREx aperture. We propagate nuclear-flux bounds from alternative extractions and host templates, uncertainty in the aperture matching, W3/W4 calibration errors, and wavelength-dependent MRS calibration covariance [44].

Supplemental Information

Special requirements. Each science/sky pair is linked as a consecutive, noninterruptible sequence to obtain a contemporaneous background [47]. Sky fields are selected using WISE and 2MASS catalogues and images, with at least $25''$ clearance from catalogued sources. Twenty-three fields lie within 1–2 arcmin of their nuclei; the crowded field around J20145928+2523010 uses a $135''$ offset. Background targets are marked extended and use four-point extended-source dithers, with no target acquisition. No fixed-date constraints, coordinated parallels or joint-facility time are requested. APT’s Visit Planner returns scheduling windows for all 48 visits.

Duplications. A coordinate-based MAST search on 2026 September 26, including archived and planned JWST observations, found no overlapping observations within $120''$ of any of the 24 science positions. No potential MIRI/MRS duplications were identified.

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